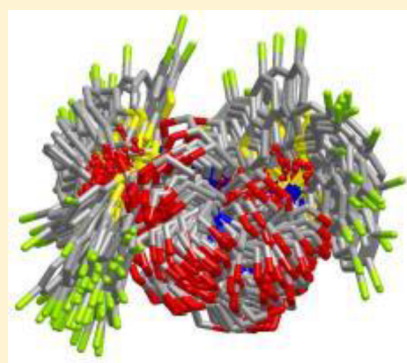


Conformation Generation: The State of the Art

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ABSTRACT: The generation of conformations for small molecules is a problem of continuing interest in cheminformatics and computational drug discovery. This review will present an overview of methods used to sample conformational space, focusing on those methods designed for organic molecules commonly of interest in drug discovery. Different approaches to both the sampling of conformational space and the scoring of conformational stability will be compared and contrasted, with an emphasis on those methods suitable for conformer sampling of large numbers of drug-like molecules. Particular attention will be devoted to the appropriate utilization of information from experimental solid-state structures in validating and evaluating the performance of these tools. The review will conclude with some areas worthy of further investigation.



KEYWORDS: Conformations, Conformers, Sampling, Validation, Molecular energetics, Stochastic, Systematic, Structural information

■ INTRODUCTION

Conformer generation continues to be a topic of great interest to computational and medicinal chemists, and a great many methods exist to sample the low energy conformational space of drug-like molecules. The goals of this review are to examine different methods of conformer generation to see how they strike a balance between quantity (size of the conformer ensemble and time required to generate it) and quality (fitness for purpose), how validation studies have been, or should be, carried out, and to provide some insight into future directions for the field. Overall, the review will be structured as a series of questions:

- The “Why?”—the purposes of conformer generation
- The “What?”—approaches to conformer generation
- The “How?”—how conformer generation tools have been evaluated and compared
- The “Where?”—unsolved problems and future directions for the field

This review will focus on tools that are fast enough to be used to process drug-like molecules in seconds rather than minutes or hours; slower methods will be treated in less detail.

The “Why?”—The Purposes of Conformer Generation.

A conformer is a distinct conformation (arrangement of atoms of a molecule) at a minimum on the molecule's potential energy surface. All the properties of that molecule are the combination of the properties of its conformers accessible at the temperature of the study. In ligand discovery, understanding the conformers that a molecule can adopt in solution greatly assists in understanding how to design conformation-dependent properties into or out of a molecule.¹ Drug-like molecules can adopt a variety of conformations, even in the solid state, as seen in collections of solid-state structures such as the Protein Data Bank (PDB)² and the Cambridge Structural Database (CSD).³ An example of conformational diversity of a single molecule in the solid state, drawn from the PDB, is shown in Figure 1. The same

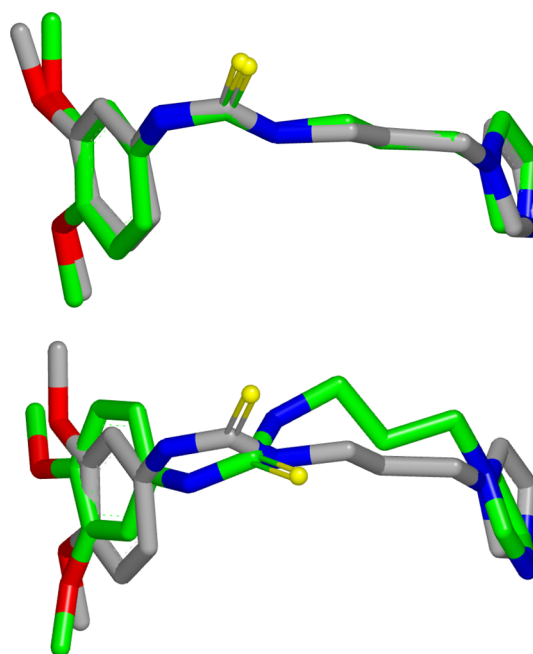


Figure 1. Conformational variability in the PDB ligand. (upper) Instances A and B from PDB structure 4f9v, RMSD = 0.22 Å. (lower) Instance A from 4f9v and instance A from 3iqe, RMSD = 2.24 Å.

ligand (1-(3,4-dimethoxyphenyl)-3-(3-imidazol-1-ylpropyl)-thiourea, three letter code PBD) is found in multiple conformations; in complex with the same protein (PDB code 4f9v) these conformations are quite similar (RMSD < 0.25 Å),

Received: April 19, 2017

Published: July 6, 2017

but the conformations taken from two different versions of the same protein (4f9v and 3pb7) are quite different (RMSD > 2 Å).

This single observation is reflective of the situation in the PDB as a whole. Figure 2 shows the distributions of pairwise RMSDs

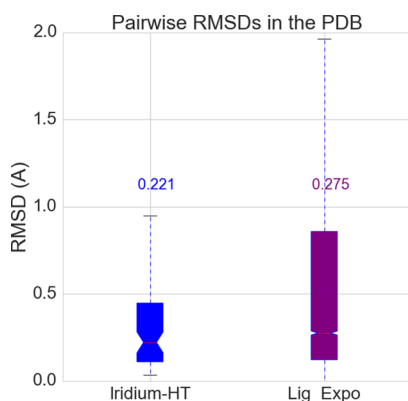


Figure 2. Conformational variability in drug-like ligand structures from the PDB. Results in blue are from a carefully curated set (Iridium-HT), results in purple are from a larger and not hand-curated set (Ligand Expo). Numbers are the median values.

of conformers of the same molecule in sets of small molecule ligands that exist in at least two different protein–ligand structures. One is a subset of the Iridium-HT database, which is a collection of carefully hand-curated and re-refined ligand structures;⁴ the other is a subset of the LigandExpo collection,⁵ where the ligands are selected automatically to be derived from good quality models and to be good fits to their local density (the selection method is described in ref 6). The subset of Iridium-HT contains 13 separate molecules, while the subset of the LigandExpo collection consists of 693 molecules. In both cases, the median pairwise RMSDs are small (the median is less than 0.3 Å), but the variance is quite large, indicating that while most molecules adopt quite similar conformations when bound to their target protein, some can adopt substantially different conformations (for methods to investigate experimental ligand conformational variability, see ref 7).

Given that drug-like molecules can exist in different conformations, even in the solid state, computing relevant conformations for them is of great importance. Those conformers provide input for several different types of calculation in lead discovery and lead optimization, including docking,⁸ shape similarity,⁹ pharmacophore searching,¹⁰ 3D-QSAR,¹¹ estimation of physicochemical properties,¹² and thermodynamic calculations.¹³ The optimal conformation sampling regime for each of these problems can be quite different. For thermodynamic calculations, a more complete sampling (of all low energy conformers) provides a better estimate of the quantity calculated: more quantity provides higher quality. In other cases, changing the conformational sampling regime has less effect. This is shown in Figure 3 for virtual screening using shape-based alignment and scoring with ROCS⁹ on the DUD data set.¹⁴ One molecule, the crystallographic ligand from the protein–ligand complexes in DUD, was used as the query in each calculation. In this experiment, increasing the conformational sampling of the database has no effect on performance past a maximum of 25 conformers per molecule. Similar findings have been made with other 3D searching methods.¹⁵ Conformer generation therefore needs to be conducted in a fashion relevant to the manner in which the conformers generated will be consumed. Simply

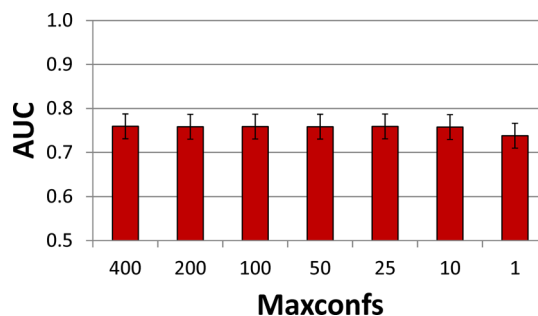


Figure 3. Virtual screening performance of ROCS on the DUD data set. Results are the median with a 95% confidence interval. Maxconfs indicates the maximum number of conformations for the database molecule generated with OMEGA.

generating more conformers does not always provide better performance: increased quantity does not necessarily guarantee increased quality.

The “What?”—Approaches to Conformer Generation.

The conformer space (bond lengths, bond angles, and torsions) for even relatively small molecules is enormous; searching for relevant conformers could therefore be time-consuming and generate a huge number of conformers (vide infra). Certain approximations have been applied to make the problem space more manageable, the most important of which is the so-called rigid rotor (RR) approximation in which bond lengths and bond angles are kept fixed. In most cases, this approximation is not harmful, especially for ring conformations,¹⁶ though in some acyclic bonds it has been shown to be problematic.¹⁷ Under the RR approximation, the available conformational space of a molecule can be easily estimated with crude heuristics;¹⁸ a molecule with 10 rotatable bonds could have more than 59 000 conformations, an overwhelmingly large number to generate, store, and, most important, process in downstream applications. However, given that only low energy conformers (usually those accessible with a reasonable probability at around room temperature) are relevant, and if reasonable bounds of minimum interconformer difference are also imposed, then the available conformational space becomes reasonably small¹⁹ and sampling in that space effectively becomes approachable.

The complementary problem to effective and relevant sampling—or generation of candidate conformations—is scoring, or ranking those conformations by some function related to their stability. Some sampling methods opt to explore as much of the low-energy conformational space of a molecule as possible, with the resulting high cost in time and number of sampled conformers required for an accurate representation of that space. Other methods choose to limit in some way the space to be searched, often by using the RR approximation, trading a (possibly small) decrease in accuracy for a (hoped for) large improvement in run-time and/or a large reduction in the number of conformations generated. In this review, conformation generators will be divided, somewhat artificially, into two broad classes based on their sampling methods, stochastic and systematic, though it should be noted that hybrid approaches are possible. For the sake of brevity, the reader is referred to the original publications for details of the tools mentioned in the subsequent discussion.

■ STOCHASTIC METHODS

The most complex and time-consuming stochastic methods for conformational sampling are those based on molecular dynamics

(MD).²⁰ MD applies Newtonian physics to study the evolution of conformation over time using, at the time of writing, a molecular mechanics force field to estimate energetics. MD attempts to sample the configurational space of a molecule widely; sampling occurs in the phase space of the problem and if sufficient sampling is achieved, then a Boltzmann-weighted ensemble of conformations can be obtained directly. This allows the calculation of any desired property of the isolated molecule (particularly useful for physicochemical property calculations). MD methods combine sampling and scoring explicitly in the calculation (i.e., the conformer score or energy is produced directly as a result of the conformer generation process). Here, the balance between quantity and quality is struck in favor of quantity of sampling, at the cost of high compute times and, usually, the number of conformers produced. Due to this high time burden, few conformation generation tools utilize classical MD and this method will not be discussed further here; recent progress in this field has been reviewed elsewhere.^{21,22}

Stochastic methods based on Monte Carlo-simulated annealing²³ (MC) are often faster than MD methods.^{24,25} Methods based on low-mode or normal-mode searching attempt to retain the sound physics of MD while requiring still less computation time.^{26–29} However, all these methods can still require many CPU minutes to produce conformations for a single molecule and are therefore unsuitable for processing even a few thousand molecules. Additionally, for MD and MC approaches (but, to a lesser degree, low-mode MD methods), the input conformation, which is required in all these methods, can bias the sampling, requiring the use of either multiple input conformations or longer sampling time. A stochastic approach that removes any possible bias arising from the input structure is distance geometry (DG). Distance geometry methods³⁰ attempt to maintain the centrality of reasonable physics (as represented by the molecular potential function used) in generating conformers, as do MD and MC methods, but are much less time-consuming. In DG, a randomly generated set of atomic coordinates (which removes both the necessity for a 3D input conformation and any bias therefrom) is refined against a set of distance constraints to generate a rough 3D conformation. This conformation is usually then optimized against a molecular mechanics force field to generate a candidate conformer. The process is repeated until sufficient conformers have been formed. A number of DG-based methods have been implemented,^{31–33} and some of them have achieved a good balance between run time and accuracy.³⁴ Run times for these methods can be in the seconds-per-molecule range, appropriate for processing large databases of molecules, with good accuracy of reproduction of solid-state conformations (*vide infra*).

In the methods discussed so far, the sampling and scoring parts of conformer generation take place in a single process (though a post-generation scoring step could be added if desired). As mentioned above, it is possible to separate these two parts explicitly, using first a sampling step then a scoring step. Methods of this class attempt to minimize calculation time through use of the RR approximation (only sampling in torsion space). The most prominent method for stochastic sampling of this possibly vast torsion space has been genetic algorithms (GAs).^{31,35–37} Setting torsions by sampling with a GA requires a 3D conformation with which to begin the process, which may be required as input³⁶ or generated *de novo* from a DG method³¹ or a library of 3D fragment conformations. GA-based methods, like DG methods, have been able to balance accuracy of reproduction, ensemble size, and calculation time well.³⁸ Aside

from GAs Monte Carlo methods has been used for torsion sampling, with reasonable results.³⁹ As an alternative to stochastic sampling in torsion space, the poling method⁴⁰ aims for stochastic sampling in energy space. This approach seemed promising but has not been widely adopted.

A general problem with stochastic methods is the possibility of nondeterministic output. The amount of sampling required to achieve a given accuracy of structure reproduction is unpredictable *a priori*. While extensive sampling with a stochastic method will often converge to provide identical output, this can increase run time substantially. An obvious possible solution to this problem is to use a systematic searching method, where the size of the problem can be predicted *a priori*.

■ SYSTEMATIC METHODS

Systematic conformer generation methods, which rely explicitly on the RR approximation, attempt to enumerate all possible (usually acyclic) torsions of a molecule.^{41,42} These methods, like GA-based searching, require a starting 3D conformation. Systematic enumeration of all possible torsion angles based on that starting conformation, the so-called brute force approach, can quickly result in a combinatorial explosion of candidate conformations that will rapidly overwhelm either the conformation generator itself (due to excessive memory requirements) or the downstream processing tools. As such, brute force systematic search methods have fallen out of favor in recent years, and rule-based variants have become much more widely used. Rule-based conformation generators limit the conformational space they explore by using a knowledge base of allowed torsion angles or allowed paths⁴³ and possibly a library of 3D fragment conformations. Limiting the sampling of torsions and flexible rings only to those specified in these knowledge bases reduces the size of the conformational space for a molecule compared to the brute force approach and allows a faster and more complete exploration of that space. A possible cost for this benefit is that rule-based conformation sampling can be coarser and more limited than the sampling available either from brute force systematic enumeration or from the stochastic methods; the hope is that judicious selection of the rules will allow the relevant parts of conformational space to be covered adequately while ignoring those parts that are not relevant. Here, in contrast to MD and MC methods, the balance between quality and quantity is struck in favor of reducing quantity (of time and usually conformers produced) while maintaining quality.

Rule-based conformer generators hold out the promise of a relevant sampling of conformational space (if the rules that are used are appropriate), consistent, reproducible output (whether that is reproducibly good or reproducibly poor can only be determined by rigorous validation), and high speed. Run-time data for a variety of different methods has recently been published⁴⁴ and results from a subset of these methods, one from each different class, is shown in Figure 4. A brief summary of the algorithms underlying the tools mentioned in Figure 4 is contained in Table 1. The data clearly show that rule-based and stochastic methods can perform similarly; of the fastest and best scaling methods, one is rule-based (OMEGA⁴⁵) and one is stochastic (ETKDG³⁴).

The knowledge base of allowed torsion angles is usually derived from analysis of torsions in solid-state structures from the PDB,⁴⁵ the CSD,^{46,47} or both, though obtaining torsion preferences from force fields or other potentials is also possible.⁴⁸ The ring conformations are either computed⁴⁹ or obtained from the CSD;⁴⁷ geometries for rings missing from the prebuilt library

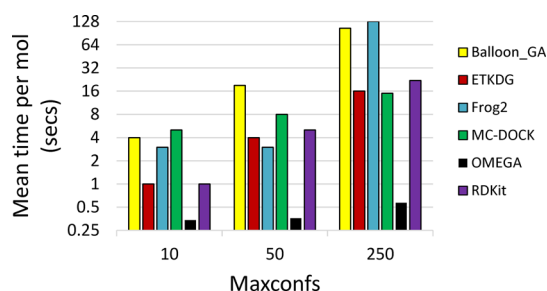


Figure 4. Mean time required to generate conformers for a set of 2912 ligands from the PDB with a set of different tools. Data for OMEGA is from the author. Maxconfs indicates the maximum number of conformers to be generated per molecule.

Table 1. Algorithmic Basis for Tools Shown in Figure 4

Tool	Type	Algorithm	Citation
Balloon_GA	stochastic	genetic algorithm	31
ETKDG	stochastic	distance geometry + knowledge base	34
Frog2	stochastic	Monte Carlo	39
MC-Dock	systematic	brute force; anchor and grow	42
OMEGA	systematic	knowledge-based; complete enumeration	45
RDKit	stochastic	distance geometry	34

are usually calculated on the fly.⁴⁶ While most of these tools assume that torsions are independent from one another, there are rule-based conformer generators that can treat pairs of torsions as dependent,^{49–51} though whether this results in a consistent improvement in performance has not been demonstrated.

As found with run time, the accuracy of rule-based and stochastic tools can be similar; data for the reproduction of a large set of high quality ligand structures from the PDB by a set of different tools are shown in Figure 5. Among the most accurate tools are a stochastic method based on DG, ETKDG,³⁴ and a rule-based method using torsion data from the PDB, OMEGA.⁴⁵

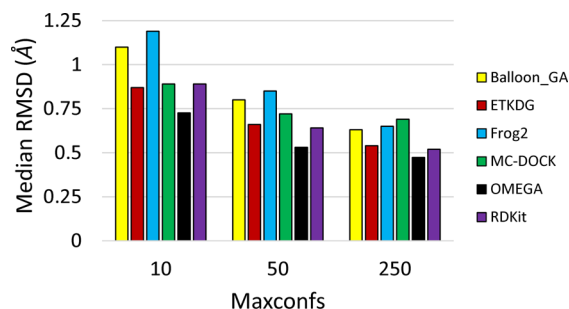


Figure 5. Median RMSD of 2912 ligands from the PDB for a set of different tools.⁴³ Data for OMEGA is from the author. Maxconfs indicates the maximum number of conformers to be generated for each molecule.

With conformation sampling achieved, either stochastically or in a rule-based manner, it remains to score and rank the conformations before they are returned to the user. There are three broad classes of approach to conformer scoring: electronic structure-based, force field-based, and knowledge- or probability-based. Electronic structure methods such as quantum mechanical (QM) or density functional theory (DFT) calculations hold out the promise of much higher accuracy in the estimation of molecular energetics than force fields,⁵² and so, conformer

generators using QM- or DFT-based scoring continue to be pursued.³⁷ However, the time requirements for QM and large basis set DFT methods are, currently, prohibitively large (many CPU minutes per conformation⁵³) and therefore cannot to be used to process even thousands of molecules without imposing a very large computational burden. Molecular mechanics force fields such as MMFF94,⁵⁴ CHARMM,⁵⁵ Tripos,⁵⁶ and OPLS⁵⁷ score conformations much more quickly than QM or DFT and are therefore very widely used in conformer scoring. In spite of their differing functional forms and methods of parametrization, these force fields all appear to perform similarly for conformer scoring.^{36,58,59} Unexpectedly, differing treatment of terms in the force field also appears to have little impact. In a number of conformer generators, the electrostatic term is omitted^{45,60} to avoid the problem of collapse to a folded conformation. In other tools, the complete force field is used,⁶¹ while in others only the vdW term is used;⁴⁹ in yet others, the vdW term is omitted³⁹ or modified.⁴⁵ Some justification for the divergence in using electrostatics in conformer scoring can be found in work⁶² showing that conformation sampling either in a model of aqueous solution or in the gas phase provides similar results when reproducing conformations found in the solid state. Sampling in the aqueous phase will give electrostatic interactions less influence on overall conformer energy due to the screening of charges on polar atoms, so neglect of the electrostatic term when scoring conformer energies is not unreasonable.

The third approach to conformer scoring attempts to capture the physics of intramolecular interactions through methods based on the relationship between frequency of a particular interaction type, probability, and energy. This method, while in its infancy for conformer scoring, is very similar in concept to the much better explored concept of potential of mean force (PMF⁶³) used in protein–ligand scoring (ref 64 and references therein). Conformer scoring methods based on distributions of experimental data from the CSD^{47,65} and the PDB⁶⁶ have been published. In Figure 6, results for reproducing the structures for a

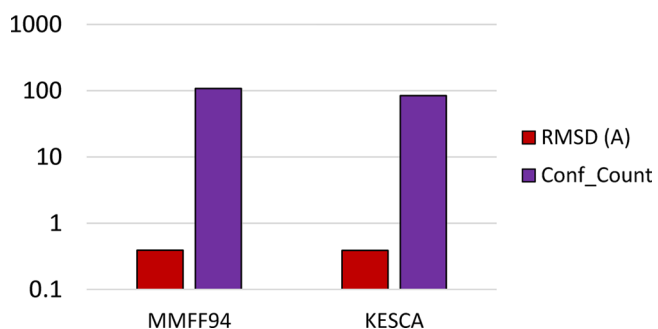


Figure 6. Reproduction of 286 structures from the CSD using two different methods of conformer scoring: a force field (MMFF94) method and a knowledge-based method (KESCA). RMSD is the median RMSD across the set; Conf_count is the median number of conformations generated.

set of 286 molecules from the CSD are shown for two methods, OMEGA,⁴⁵ which uses a force field for scoring (MMFF94), and the MT method,⁶⁶ which uses a PMF-like scoring function (KESCA) derived from information from protein–ligand complexes in the PDB.⁶⁴ The results in terms of reproduction and the number of conformations generated to acquire that level of reproduction are very similar; the two tools have a similar balance of quantity and quality.

The large diversity of potential functions and the components of these functions that have been used in conformer generators is shown in Table 2 (this list is not exhaustive but is merely illustrative).

Table 2. Potential Functions and Components of That Function Used in Some Conformer Generators^a

Tool	Potential	Components	Optimization
Balloon_GA	MMFF94	bend, stretch—bend, vdW	yes
Confab	MMFF94	all	no
ConfGen	OPLS	torsion and vdW	yes
Cyndi	Tripos or MMFF94	vdW and torsion	optional (no Coulombic)
ETKDG	CSD-derived torsion potentials	all	yes
Frog2	MMFF94	vdW only	optional
MC-Dock	Amber	vdw and Coulomb only	no
OMEGA	MMFF94	no attractive vdW No Coulombic	no

^avan der Waals.

In sum, for fast conformer generators, there seems no general advantage to a particular method of sampling (rule-based or stochastic) or to scoring with a particular potential function (molecular mechanics force field, as a whole or in part, or knowledge-based potential). Electronic structure methods of scoring are still too slow for widespread use but are vital for success in accurately predicting some physical properties.⁵³ Knowledge-based methods, particularly for scoring, are still in their infancy but hold out much promise for the future, *vide infra*. With many possible solutions available, reliably selecting among them requires rigorous validation.

The “How”—Validating Conformation Sampling Methods. The goal of validating a tool is to make predictions—predictions of the performance of that tool on systems it has not yet been tested against. These predictions are consumed by users when selecting a tool for a given task, and also by developers to help guide future improvements to the tool.⁶⁷ Therefore, validation data sets and methods should be able to provide both reliable predictions of future performance and guidance for future improvements. Two approaches to validating conformational sampling tools are assessing sampling of conformational space around experimentally identified conformations and measuring the diversity of the conformers generated. The first of these methods is used in almost every publication on conformational sampling tools, perhaps in part because it is a straightforward experiment to conduct (at least on the surface, *vide infra*). The second is used much less often and is done differently in almost every publication that attempts it; as with other aspects of diversity in cheminformatics, asking for diversity is much easier than measuring it and knowing whether or not it has been achieved.

■ ACCURACY OF REPRODUCTION

There are many ways to usefully approximate a particular conformer of a molecule¹⁷ and, at a minimum, only one is required for a downstream method to be successful (though more will raise the probability of success). The difficulty of reproduction, predictably, increases with molecular flexibility⁶¹ and so, for highly flexible molecules, a close reproduction of a given reference conformation is still more likely to be a matter of

luck or overtraining than a consistent result arising from a well-performing tool.¹⁸ Success in reproducing a solid-state conformation is usually calculated based on the closest matching conformer generated, where the match between the calculated conformer and the one derived from experiment is almost always measured by RMSD of the heavy atoms of the molecule. It is common to apply RMSD cutoffs for success in reproducing a target conformation (<0.5 Å, <1 Å, <1.5 Å, or <2.0 Å are often used in conformer generation studies), as well as using sample statistics like mean and median of the overall results; it is arguable that success rates, particularly at high accuracy, are the most important metric. In spite of its popularity, there are well-known problems with RMSD as a measure of deviation.^{45,68} A major difficulty is the scaling of RMSD with molecular size (constant RMSD indicates less deviation as molecular size increases). Another is that, after a certain point, higher RMSD simply indicates that the prediction is wrong (RMSDs of 6 or 8 Å simply indicate functionally equivalent poor performance).⁶⁸ Also, RMSD is bounded only below, by 0, with no obvious method to compute the upper bound in any particular case. Consequently, other measures have been sought,⁶⁹ with those that are bounded by construction being the most useful as they have the same meaning for molecules of different sizes (e.g., similarity in torsion fingerprints³⁵ or similarity in shape,⁷⁰ measured by Tanimoto coefficients, which are bounded by 0 and 1).

In the great majority of cases, data sets for validating conformer generators by this method are drawn from the PDB, as reproduction of protein-bound conformations is often the key goal for a conformer generation tool. Since predictions from these data sets are of great interest to developers and users of conformer generators, it is relevant here to make some comments on the method by which the ligand conformations being reproduced are obtained. The atom coordinates in X-ray protein–ligand structures are models⁴ generated to explain the data (structure factors); they are not data themselves. It is possible to estimate the quality of the model (how well the atom coordinates in the model reproduce the experimental electron density) using a number of quantitative metrics^{4,45} and also by simple visual inspection.⁷¹ Unfortunately, many evaluations of the accuracy of reproducing ligand conformations from the PDB have used essentially no quality criteria to select the ligand structures to be reproduced. In by far the majority of published validation studies where quality criteria have been used to select ligand structures, resolution of the parent structure has been used to select suitable ligands (purely as examples, see refs 24 and 72). Resolution is a measure of the quantity of data used to develop a model, not a measure of how well the model fits the data, or the model quality, and therefore it should not be used as a metric of model quality.^{4,73} A much better metric of overall model quality is the average error in the coordinates of a model structure (its diffraction coordinate precision index or DPI⁷⁴), but this is rarely used in ligand selection.⁴⁵ Unfortunately, in the area of conformer prediction, unlike physical property prediction, only a few reports pay attention to incorporating this experimental error in the structures being predicted into their analysis.^{45,67,75,76}

Fortunately, two ligand data sets paying close attention to model quality are now available: the Iridium data set⁴ and the larger and more recent Platinum data set.⁴⁴ Structures in the Iridium data set have been selected using DPI as a component of assessing model quality. In Iridium, each model was carefully manually checked for its representation, and each ligand was re-fit

to its electron density, which compensates for the high error rate in ligand structures in the PDB.⁷⁷ However, the small size of the data set (121 molecules) gives low statistical power (a factor almost always overlooked in experimental design in CADD⁶⁷) and covers only a small part of known chemical space. The much larger Platinum data set (4626 molecules) has higher statistical power and coverage of chemical space, but each ligand conformation is taken as is from the PDB without individual quality checking, refitting, or assessment of the model's DPI. One or, preferably, both of these data sets should become the standard for future comparisons and validations in this area.

Another vexing issue in the use of ligand conformations from the solid state for validating conformer generators, particularly those from the PDB, is (apparent) ligand strain. Any energy put into straining a ligand away from its lowest energy conformer is energy that cannot be realized in binding.³⁵ High affinity ligands are therefore unlikely to have even moderate levels of conformational strain. Since molecular properties are the weighted sum of the properties of individual conformers, it is unlikely that a conformer with genuine strain of much over 2 kcal/mol will influence molecular properties in any meaningful way at room temperature.⁷⁸ There have been a number of publications attempting to estimate strain energy in ligands bound to proteins; all of them have focused on enthalpic strain, based on differences in enthalpy between a notional (not experimentally observable) global energy minimum conformation in some potential function and the nearest local minimum to the conformation found in the solid state. Early attempts to estimate strain used force fields^{79,80} and found strain enthalpies of <10 kcal/mol for the great majority of ligands. High level electronic structure studies of ligands in the PDB have in some cases found higher levels of strain (>10 kcal/mol) to be strikingly frequent,⁷⁶ while other studies at similar levels of theory⁸¹ have reported much lower enthalpic strain (up to 2 kcal/mol). It should be noted that high apparent enthalpic strain is often found in molecules that can make intramolecular hydrogen bonds, which complicates the analysis of ligand strain at a large scale.⁶⁶ Another, more frequently found, source of high apparent strain lies in the extremely simple force field contained in the software usually used in fitting small molecule structures to electron density.^{4,82} It has been shown, unsurprisingly, that when a better representation of small molecule physics (a more sophisticated force field or electronic structure method) is used to fit a ligand, the resulting structural models are equivalent in quality but lower in strain compared to those generated using the traditional approach.^{83,84} It is to be hoped that these more physically accurate methods of fitting ligands to density will become more widely adopted in the crystallographic community.⁸⁵ However, until substantial effort is put into refitting existing protein-bound ligand structures using these methods (see, for example, PDB-REDO⁸⁶), attempting to assess the actual level of strain induced upon binding a ligand to a protein across a wide variety of ligands will remain problematic.

Given all the difficulties with ligand conformations derived from the PDB, some investigators have turned to the CSD for reliable structural models for testing their tools.^{34,45,66,72} CSD structures are usually accurate⁷⁷ and errors in the coordinates of structures from the CSD are usually vanishingly low compared to the errors in the predictions made by conformer generators, allowing greater precision for comparing performance. The differences between solid-state structures of the same molecule in the PDB and the CSD, usually perceived as being due to the influence of crystal packing forces on conformations in the CSD,

have in fact been shown mostly to be due to the coincidence of crystallographic and molecular symmetry,⁸⁷ not genuine differences in the conformations themselves. Therefore, the CSD presents a rich mine of data for validating conformer generators aimed at reproducing conformations found in a protein–ligand complex, *vide infra*, and should be more widely used in future work.

■ CONFORMER DIVERSITY

There have been several approaches to describing the diversity of an ensemble of conformers in accessible conformational space: comparison to an estimation of the complete ensemble accessible to a given molecule,^{19,88} information entropy,⁸⁹ the size and distance between clusters of similar conformers,⁹⁰ and distributions of interconformer RMSDs.³⁵ While intraset conformer diversity is almost always controlled using a minimum interconformer RMSD during the generation process, the community has not yet concurred on a standard measure for the diversity between sets of conformers. Since conformer diversity is difficult to measure in a reliable way, it has proven even more difficult to represent conformer diversity and so few attempts to visualize conformational space have been attempted. The contrast with assessing the diversity of collections of molecules at the graph level is marked; a number of approaches to calculating graph diversity exist and the representation of chemical space has been quite widely investigated.⁹¹ Notable early attempts to visualize the occupied conformational space of a molecule include those by Kirchmair⁷² and Loferer,⁹² but a good deal more work remains to be done in this area.

The “Where?”—Future Directions for the Field. There are some clear directions for future improvement in conformer generation tools, a few of which are outlined in this section.

■ USE OF SOLID-STATE INFORMATION

In the future a major direction for conformer generators will be the incorporation of more and more experimentally derived structural knowledge from both the PDB and the CSD, particularly the CSD. As noted above, the CSD provides a wide range of highly reliable solid-state structures against which to validate conformer generators. More importantly, knowledge derived from the CSD can be applied to several problems in conformer generation relevant to the protein-bound state;³ property distribution data from the CSD can improve the accuracy of predicting bound conformations found in the PDB, both in terms of conformation generation⁹³ and pose scoring.⁹⁴ Knowledge-based approaches to conformation generation using torsion angle distributions,^{35,47} fragment geometry,^{47,50} and even pseudo-energies⁶⁵ derived from the CSD have been shown to be generally accurate. CSD-derived data has been applied to the validation of bond lengths,¹⁷ torsion angles,⁹⁵ and ring conformations¹⁶ in computed or experimentally obtained (crystallographic) conformations.

As mentioned above, most of the deviations between conformational properties in the CSD and the PDB arise due to issues with crystal symmetry.^{87,96} In most of the remaining cases, the differences arise due to a misfit PDB ligand structure; refitting the ligand using methods that better deal with small-molecule ligand energies produces ligand conformations in better agreement with CSD data.⁸⁵ The oft noted lack of concordance between PDB and CSD torsion frequency distributions for some functional groups is therefore not “a direct consequence of using typical force fields for fitting

coordinates to electron densities,” as has been suggested.⁹⁵ Historically, for structures deposited in the PDB, typical molecular mechanics force fields have not commonly been used to fit ligand structures to electron density but, highly simplified molecular potential functions⁹⁷ (which are less physically reasonable representations of molecular geometry and energetics even than force fields). When molecular mechanics force fields are used to fit ligands to density, concordance between the PDB and the CSD geometries improves.⁸⁴ A reasonable objection to the use of CSD data to assess conformer stability or probability over theoretical methods like QM and force fields is that in cases of novel or poorly explored chemistry, the CSD will not contain enough data to make a useful prediction of conformational properties or probabilities.⁷⁸ When dealing with chemistry that is not well represented in the CSD, judgments about the probability of a given conformation must be made with great caution: the absence of evidence is not evidence of absence. According to recent work,⁹⁵ approximately 15% of acyclic rotatable bonds have insufficient occurrences in the CSD to make useful deductions about their torsion preferences. It is also worth recalling that, even for cases that are apparently well-represented in the CSD, any probability distribution extracted from experimental data should be carefully manually analyzed before use. In those cases where available experimental data is unreliable, sparse or lacking altogether, the only recourse is to use computed data. It has been shown a number of times that CSD torsion distributions are often well reproduced by calculations at even relatively low-level quantum mechanical methods,^{98,99} so QM results should be an acceptable stopgap until sufficient experimental data have been accumulated.¹ Given the pace of growth of the CSD, which doubles in size every eight years, the number of cases for which this will be required will likely decline quickly. It is to be hoped that the increase in the number of high-quality ligand structures in the PDB will also contribute substantially to the knowledge base of small molecule conformational preferences.

■ MACROCYCLES

The unusual physicochemical property space of macrocycles has made them interesting targets in a variety of drug discovery contexts.¹⁰⁰ Due to their large size and flexibility, macrocycles are a challenging target for conformer generation,⁵⁸ but understanding their conformational properties is key to understanding their unusual properties. Some existing tools for conformer sampling discussed here are not appropriate for macrocyclic molecules (torsion driving methods, either stochastic or rule-based), while others are very slow (MD and related methods), so different approaches will be required. Two newly developed tools with original algorithms have shown promise in their initial tests on this problem,^{101,102} but it is clear that much further work in this area is required.

■ CONFORMER ENTROPY

As discussed above, the questions of how wide a range of energy should be allowed when computing conformers or the likely range of strain energy in a bound ligand have exercised the community for some time. A key flaw in all the studies conducted to date on ligand strain is that they only attempt to assess ligand enthalpic strain and measure that strain with respect to an inappropriate ground state—the conformation with the lowest strain enthalpy. It is often stated¹⁰³ that the global minimum is the reference state for calculating pose stability or strain energy

and an enormous amount of effort is sometimes expended to find it,³⁷ but this effort is expended in pursuit of the wrong goal. The true, relevant ground state for strain estimation is not the global minimum but the entire ensemble of low-energy conformations in aqueous solution,⁷¹ the environment in which all relevant binding phenomena are measured. As such, a useful estimate of conformational strain must take into account not only enthalpic contributions to conformer stability but entropic ones as well, which requires that each conformer not be considered in isolation but with respect to its low energy ensemble in solution. The entropic contributions to conformer stability are clearly much more difficult to estimate than enthalpic terms, which may well be why this aspect of conformational strain has largely been ignored. However, there has been recent progress in this direction,^{13,26} and it is hoped that further investigations will improve the community's understanding of the contributions to entropy to ligand energetics.

It is clear from the forgoing discussion that solutions offering the ideal balance of quantity and quality are not yet available for all types of problems in conformer generation. Some time has elapsed since a review in this area asserted that the problem of conformer sampling is essentially solved for “small, drug-like molecules”¹⁰³ and that the scoring of conformers was the major remaining problem. It is this author's view that this is, regrettably, still the case. Advances in scoring on two fronts may provide good solutions: physics-based solutions that include better estimation of both enthalpic terms and entropic terms and knowledge-based solutions constructed on reliable structural data (be that from the PDB or the CSD). While adequate solutions exist for conformation sampling and scoring for many traditionally drug-like molecules, increasing interest in developing therapeutic compounds that are not small, drug-like molecules (in particular, macrocycles) continues to put pressure on the existing methods. It is hoped that the community will respond to this pressure by continuing to develop innovative and effective solutions to this problem of ongoing interest.

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Notes

The author declares the following competing financial interest(s): P.C.D.H. works for OpenEye Scientific, the vendor for one of the tools mentioned in this review (OMEGA).

■ ACKNOWLEDGMENTS

Prof. Gerard Pujadas is thanked for assistance in generating the LigandExpo-derived ligand dataset. Ms. Annie Lux is thanked for proofreading the manuscript. The anonymous reviewers are thanked as their insights contributed substantially to improving this paper.

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